L. M. Fernandes-Oliveira

MSc student in Statistics · Production Engineer

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PROFILE

MSc student in Statistics at UFPE (Brazil) and production engineer. I work with data and process analysis, statistical modeling and computational methods, and I am deepening my knowledge of machine learning. I come from Production Engineering, where I first met statistics on the practical side: monitoring processes and studies in plastics and metalworking companies.

EDUCATION

2026 – 2027 MSc in Statistics (in progress) · UFPE, Recife, Brazil

2025 BSc in Production Engineering · UFPE, Agreste Campus, Brazil

PUBLICATIONS

- **L. M. F. de Oliveira** and R. J. Cintra. Uma aproximação escalada para DCT de comprimento 16 [A scaled approximation for the length-16 DCT]. *Proceeding Series of the Brazilian Society of Computational and Applied Mathematics*, v. 11, n. 1, 2025. Presented at the XLIII CNMAC, Porto de Galinhas (PE), Brazil, 2024. In Portuguese.

SKILLS

- **Search algorithms, heuristics and metaheuristics.** Exploring large solution spaces systematically or approximately, balancing the quality of the answer against the available time and computational cost. Combinatorics.jl, Optim.jl
- **Mathematical modeling and problem definition.** Understanding the context, identifying what needs to be decided or explained, and translating it into a model with clear variables, constraints and objectives that guides the choice of method. JuMP, SciPy
- **Computer vision and image processing.** Extracting, representing and compressing visual information, adapting the technique to each application and assessing results with objective quality metrics. OpenCV, scikit-image, Images.jl
- **Fast algorithms and low-cost processes.** Making algorithms and processes more efficient by mapping where time, energy or resources are spent and proposing simplifications that cut costs without compromising quality. NumPy, LinearAlgebra
- **Statistics.** Statistical modeling, inference and multivariate analysis, using simulation and resampling to evaluate methods and quantify the uncertainty of results. pandas, Matplotlib, DataFrames.jl, Plots.jl, MASS
- **Machine learning.** Classification and regression models, with train/test splits, validation and objective model comparison. scikit-learn, PyTorch, Flux.jl, glmnet, e1071

PROJECTS (MSC COURSEWORK)

- **Spam detection in e-mails.** KNN, logistic regression, LDA and QDA on the Spambase data; logistic regression reached 91.6% accuracy and an AUC of 0.972.
- **Regularization in polynomial regression.** Ridge, Lasso and Elastic Net on high-degree polynomials, with a sensitivity analysis to noise.
- **Statistics for circular data.** Uniformity tests, von Mises fitting, bootstrap and Monte Carlo on turtle and ant movement directions.
- **Financial well-being index.** Principal components on a financial well-being scale; the first component explained 56.1% of the variance.
- **Classification methods: a comparative study.** LDA, QDA, logistic regression, KNN and RDA on the Iris data and in simulations, compared with McNemar's test.
- **Power of Hotelling's T^2 test.** Simulation of the test's power for samples of 30 to 1,000 observations and different mean shifts.

LANGUAGES AND TOOLS

Julia · Python · R · MATLAB · C · Excel